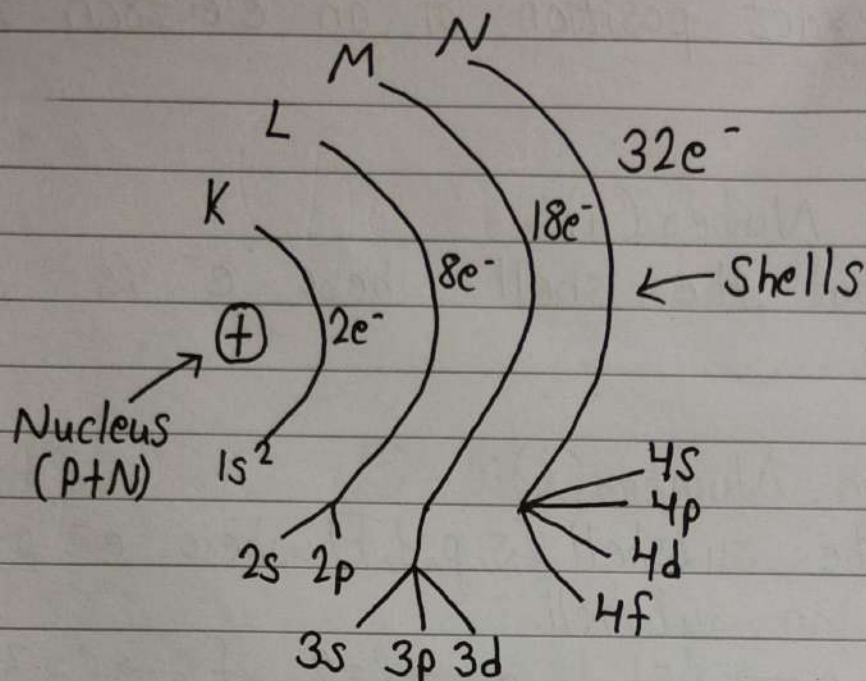


Module I: Atomic Structure & Chemical Bonding

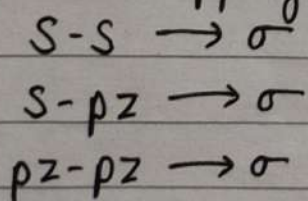


How e^- fill in atom

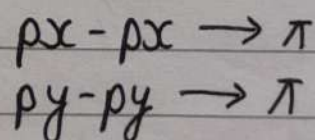
Shell	Sub-shell	Orbital
K	s	$\square$ $2e^-$
L	p	$\square \square \square$ $6e^-$
M	d	$\square \square \square \square$ $10e^-$
N	f	$\square \square \square \square \square \square$ $14e^-$

Bonds (Orbital Overlapping)

Axial overlapping (σ bond)



Non-axial (sidewise)



Note: σ -bonds are stronger than π -bonds.

Quantum Numbers

→ They give the exact position of an electron in an atom.

1. Principal Quantum Number (n):

Gives info about the shell where e^- is present. $n = 1, 2, 3, \dots$

2. Azimuthal Quantum Number (l):

Tells about the subshell (s, p, d, f) where e^- present. Values depend on subshell.

$$s \rightarrow l=0 \quad p \rightarrow l=1 \quad d \rightarrow l=2 \quad f \rightarrow l=3$$

3. Magnetic Quantum Number (m_l):

Tells about the orbital where e^- present. Values: $m_l = -l$ to $+l$

4. Spin Quantum Number (s or m_s):

Tells the direction of e^- .

Values:

$$s = +\frac{1}{2} \quad [\text{Spin clockwise } (\uparrow)]$$

$$s = -\frac{1}{2} \quad [\text{Spin anti-clockwise } (\downarrow)]$$

Note: $l < n$ always.

Find Quantum Numbers of following

① $4d^2$

$$n = 4$$

$$l = 2$$

$$m_l = -1$$

$$s = +1/2$$

② $4f''$

$$n = 4$$

$$l = 3$$

$$m_l = 0$$

$$s = -1/2$$

③ $5s^2$

$$n = 5$$

$$l = 0$$

$$m_l = 0$$

$$s = -1/2$$

④ $6d^7$

$$n = 6$$

$$l = 2$$

$$m_l = -1$$

$$s = -1/2$$

⑤ $5f^{10}$

$$n = 5$$

$$l = 3$$

$$m_l = +1$$

$$s = +1/2$$

⑥ $4f^{14}$

$$n = 4$$

$$l = 3$$

$$m_l = +3$$

$$s = -1/2$$

⑦ $6p^4$

-1	0	+1
↑↓	↑	↑

$$n = 6$$

$$l = 1$$

$$m_l = -1$$

$$s = -1/2$$

Write configuration

① $n = 2$ $l = 1$ $m_l = +1$ $s = -1/2$

-1	0	+1
↑↓	↑↓	↑↓

 $\rightarrow 2p^6$

② $n = 3$ $l = 2$ $m_l = 0$ $s = -1/2$

-2	-1	0	+1	+2
↑↓	↑↓	↑↓	↑	↑

 $\rightarrow 3d^8$

Molecular Orbital Theory

Quantum mechanics model that describes chemical bonding by combining atomic orbitals to form new, delocalized molecular orbitals that span the entire molecule.

(i) Same atom combine $\rightarrow$ Homo-nuclear diatomic molecule.
Diff. atom combine $\rightarrow$ Hetero-nuclear diatomic molecule.

(ii) Bonding orbital (σ, π) $\rightarrow$ Lower energy, stable.
Anti-bonding orbital (σ^*, π^*) $\rightarrow$ Higher energy, unstable.

(iii) Bond Order (B.O.) = $\frac{N_b - N_a}{2}$

where, N_b is no. of e^- in bonding orbitals.

& N_a is no. of e^- in anti-bonding orbitals.

Higher B.O. $\rightarrow$ Stronger bond $\rightarrow$ Shorter bond length.
If B.O. = 0 $\rightarrow$ Molecule does not exist.

(iv) Magnetic Nature:

If unpaired electrons $\rightarrow$ Paramagnetic.

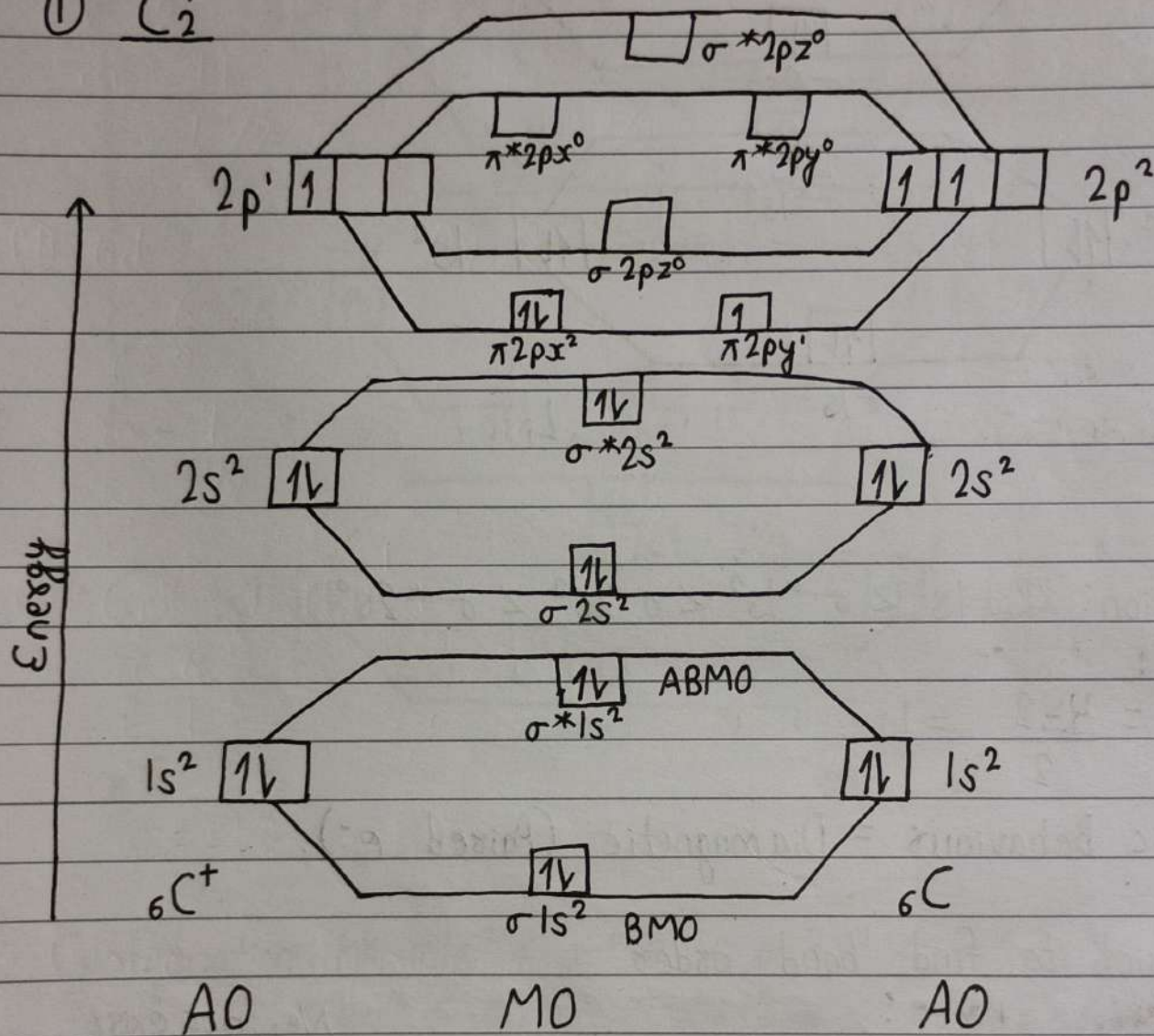
If paired electrons $\rightarrow$ Diamagnetic.

(v)	e^- 18	14 or less e^-	15 or more e^-
		1	1
		2	2
		1	2
		2	5
		4	

Homo-nuclear Diatomic Molecule

Molecules formed by two identical atoms.

① C_2^+

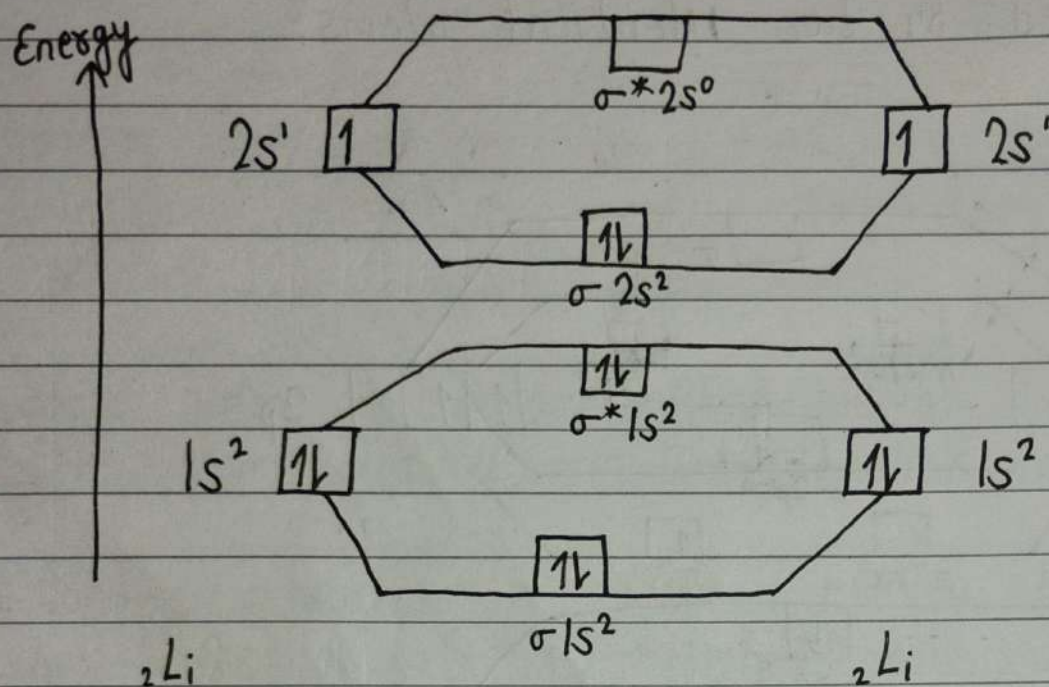


Configuration: $KK \sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \pi 2p_x^2 = \pi 2p_y^2 < \sigma 2p_z^0 < \pi^* 2p_x^0 = \pi^* 2p_y^0 < \sigma^* 2p_z^0$.

$$\text{Bond Order} = \frac{N_b - N_a}{2} = \frac{7 - 4}{2} = 1.5$$

Magnetic behaviour = Paramagnetic (Unpaired e^-).

② Li_2



Configuration: $KK \sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^0$

$$B.O. = \frac{4-2}{2} = 1$$

Magnetic Behaviour = Diamagnetic (Paired e^-).

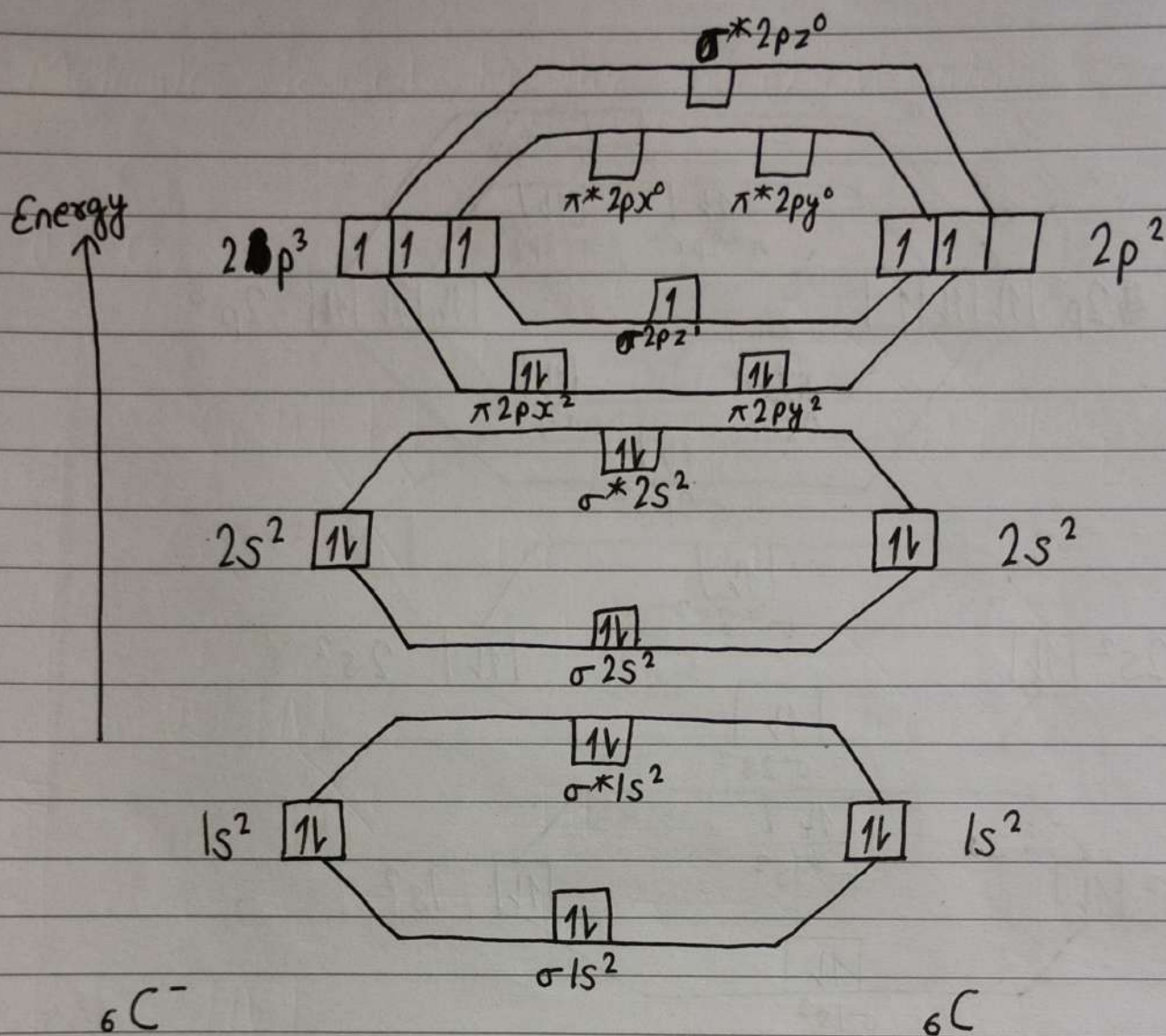
Short-Trick to find bond order

Be ₂ , Not exist						Ne ₂ , not exist							
8	9	10	11	12	13	14	15	16	17	18	19	20	← No. of e ⁻
0	0.5	1.0	1.5	2.0	2.5	3.0	2.5	2.0	1.5	1.0	0.5	0	← B.O.

0.5/1.5/2.5 $\Rightarrow$ Paramagnetic.

1.0/2.0/3.0 $\Rightarrow$ Diamagnetic.

③ C_2^-

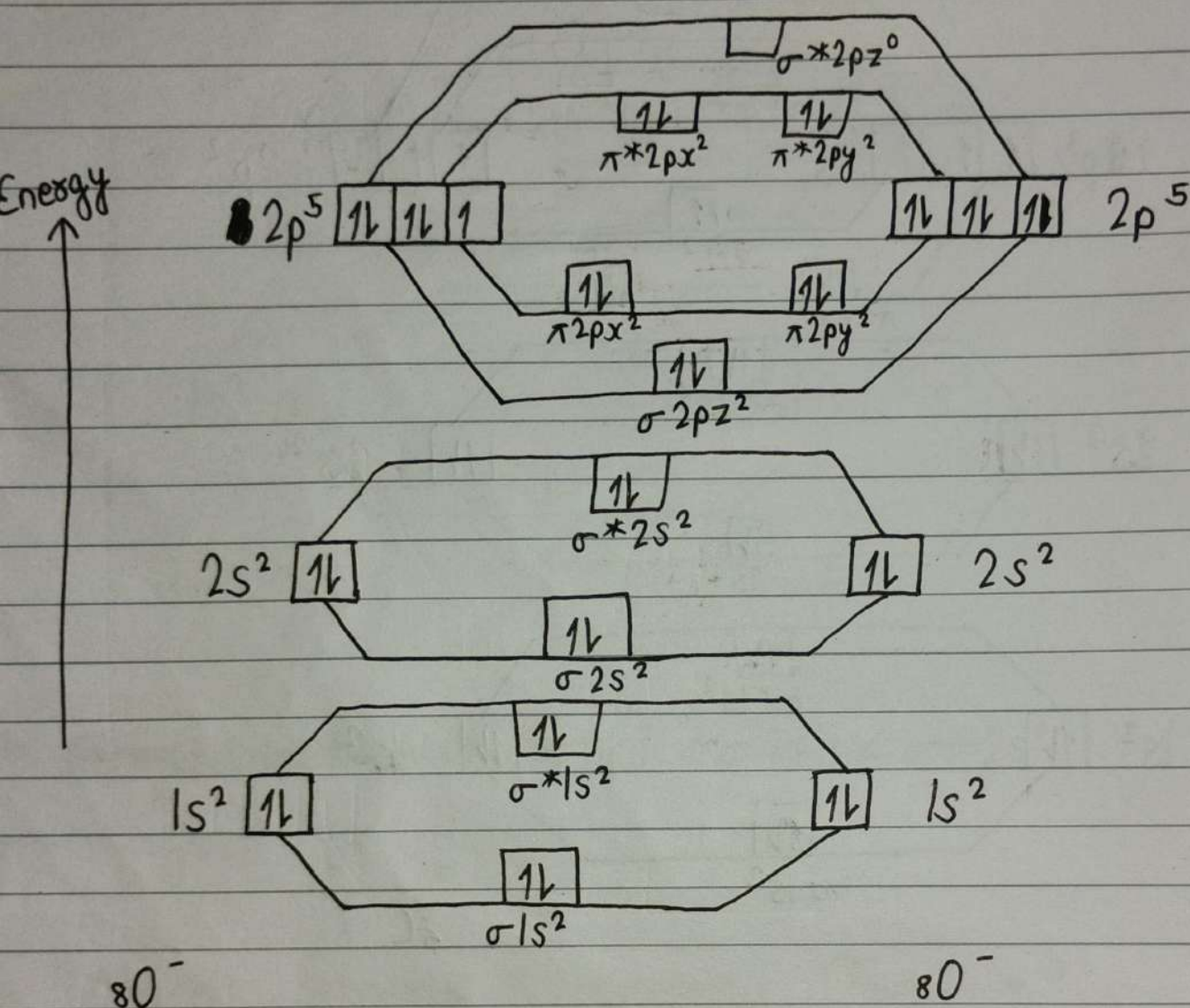


Configuration: $KK \sigma_{1s}^2 < \sigma_{1s}^{*2} < \sigma_{2s}^2 < \sigma_{2s}^{*2} < \pi_{2p_x}^2 = \pi_{2p_y}^2 < \sigma_{2p_z}^1 < \pi_{2p_x}^{*0} < \pi_{2p_y}^{*0} < \sigma_{2p_z}^{*0}$

$$B.O. = \frac{9 - 4}{2} = 2.5$$

Magnetic behaviour = Paramagnetic.

④ O_2^{2-}

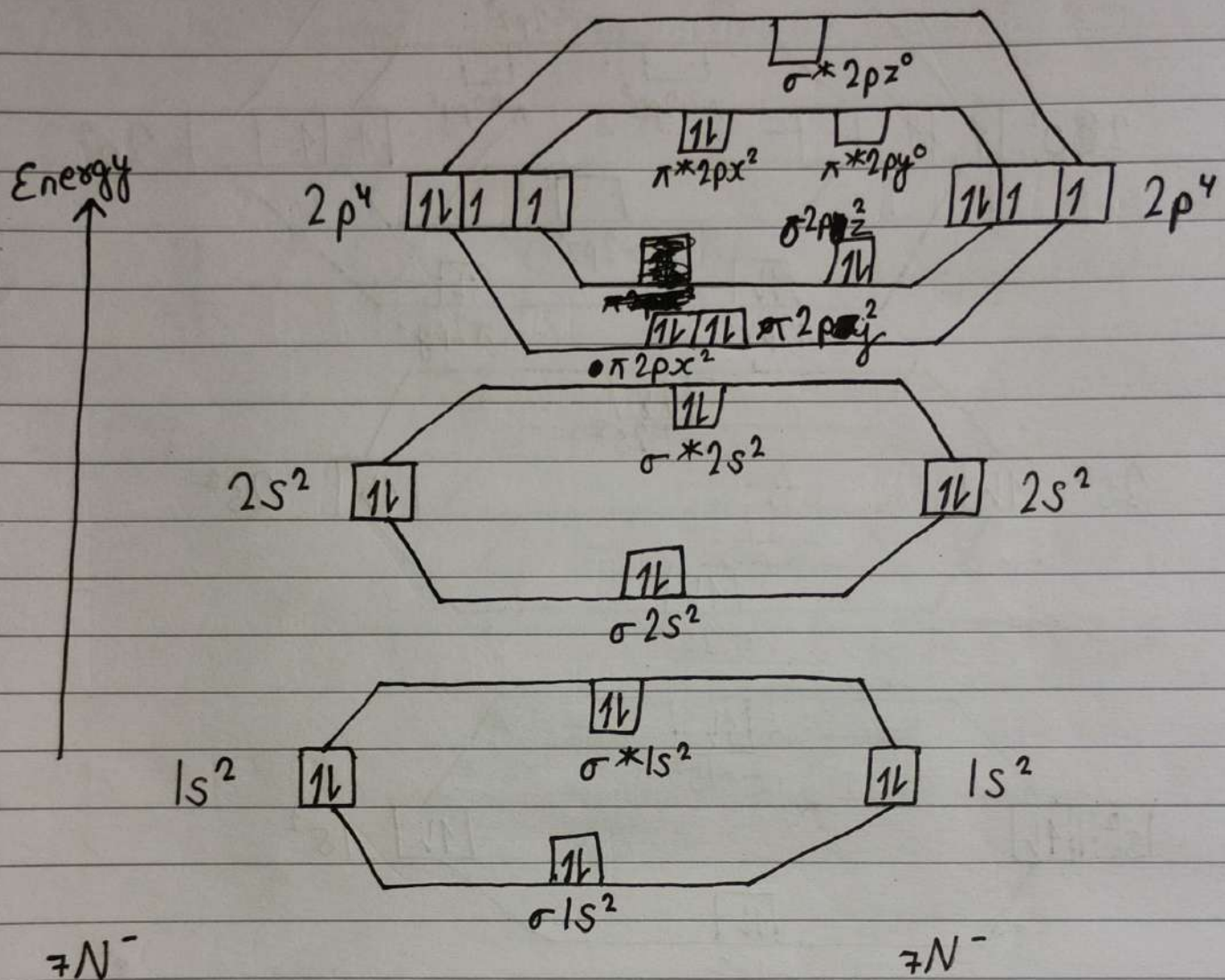


Configuration: KK $\sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \sigma 2pz^2 < \pi 2px^2 = \pi 2py^2 < \pi^* 2px^2 = \pi^* 2py^2 < \sigma^* 2pz^0$.

$$B.O. = \frac{10 - 8}{2} = 1$$

Magnetic behaviour = Diamagnetic.

⑤ N_2^{2-}

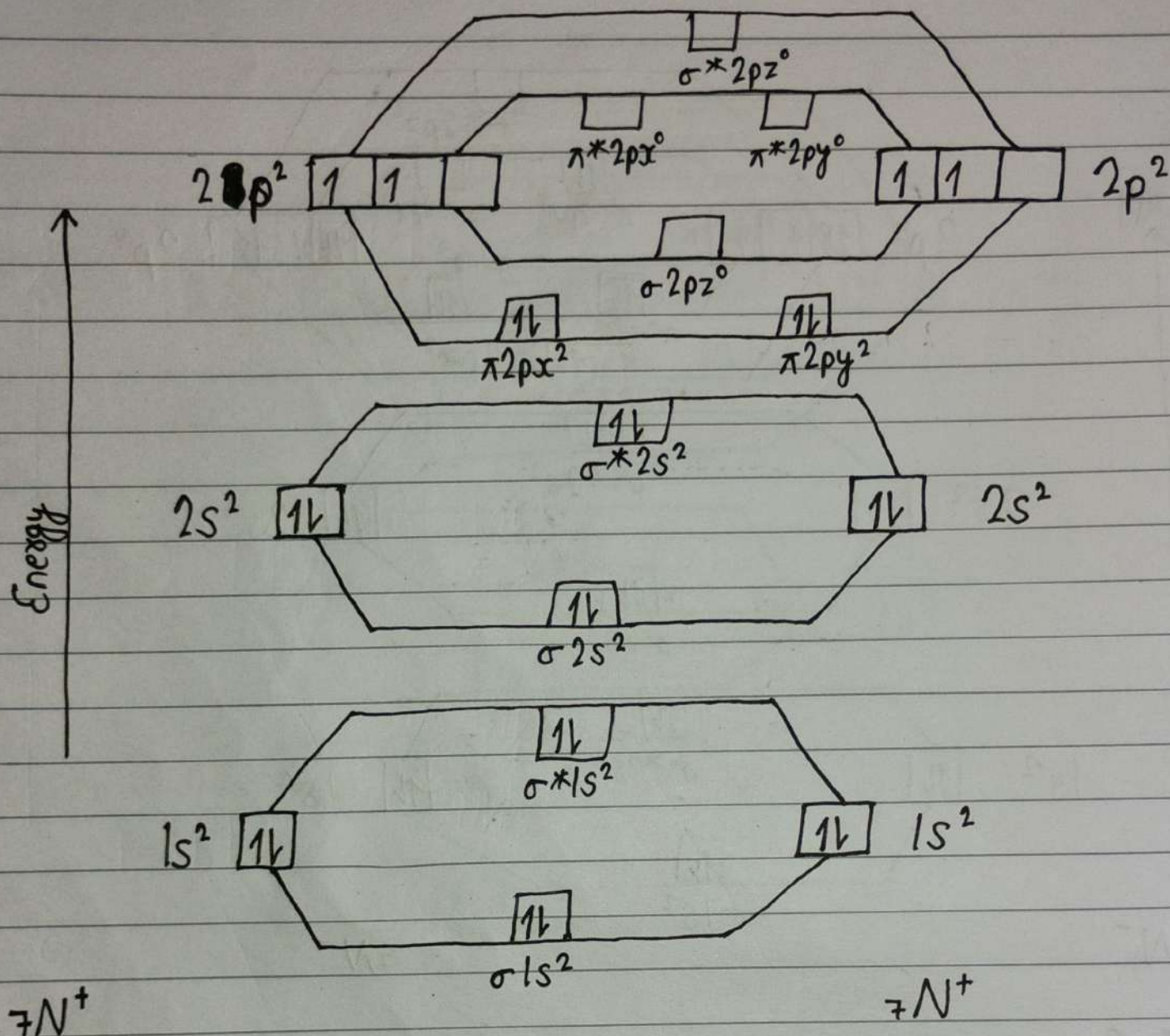


Configuration: $KK \sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \pi 2p_x^2 = \pi 2p_y^2 < \sigma 2p_z^2 < \pi^* 2p_x^2 = \pi^* 2p_y^2 < \sigma^* 2p_z^2$

$$B.O. = \frac{10 - 6}{2} = 2$$

Magnetic behaviour = Diamagnetic.

⑥ N_2^{2+}



Configuration: $KK \sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \pi 2p_x^2 = \pi 2p_y^2 < \sigma 2p_z^0 < \pi^* 2p_x^0 = \pi^* 2p_y^0 < \sigma^* 2p_z^0$.

$$B.O. = \frac{8-4}{2} = 2$$

Magnetic behaviour = Diamagnetic.

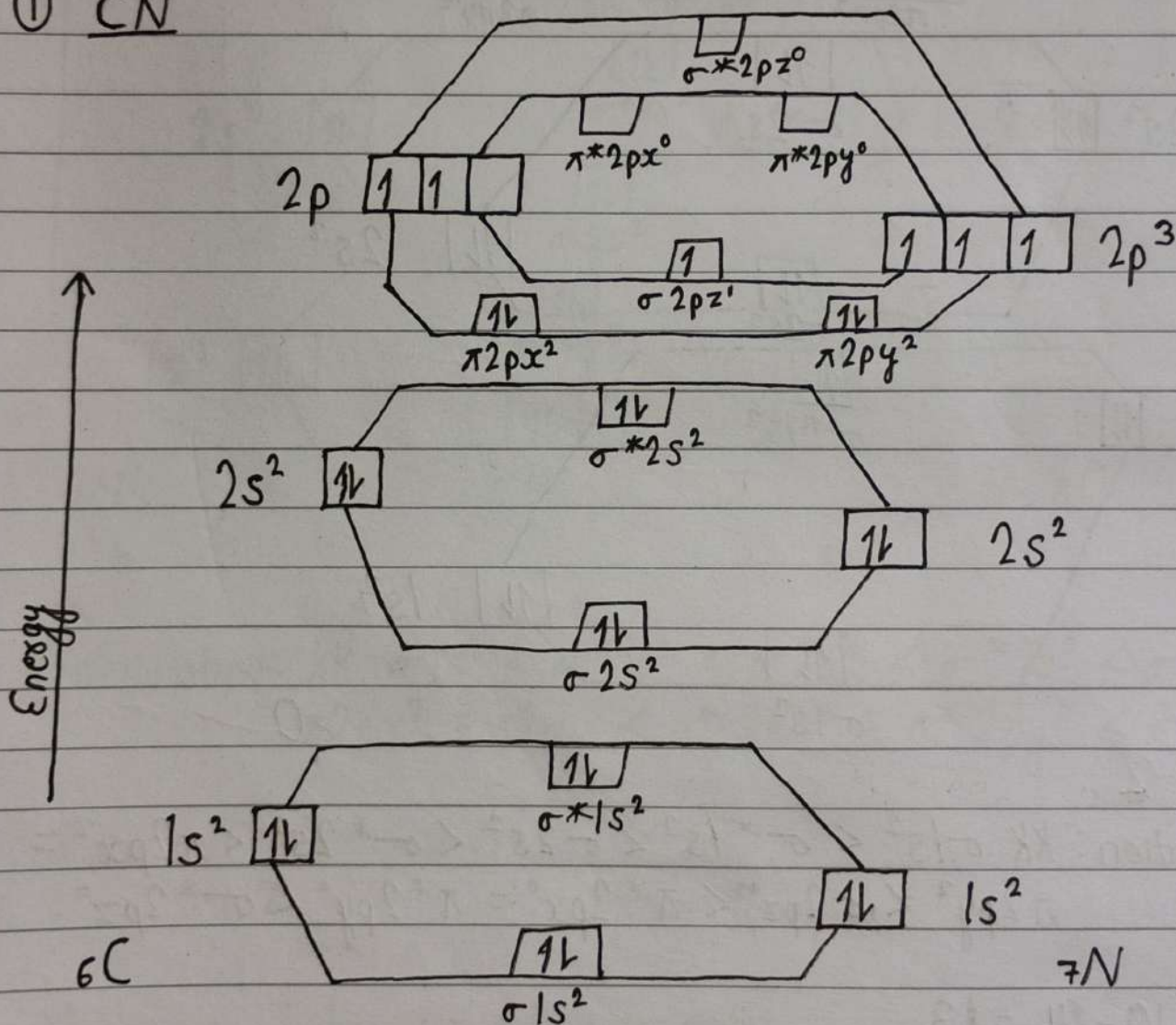
Hetero-nuclear diatomic molecules

Molecules formed by two different atoms.

Electronegativity:

H	B	C	N	O	F
2.1	2.0	2.5	3.0	3.5	4.0

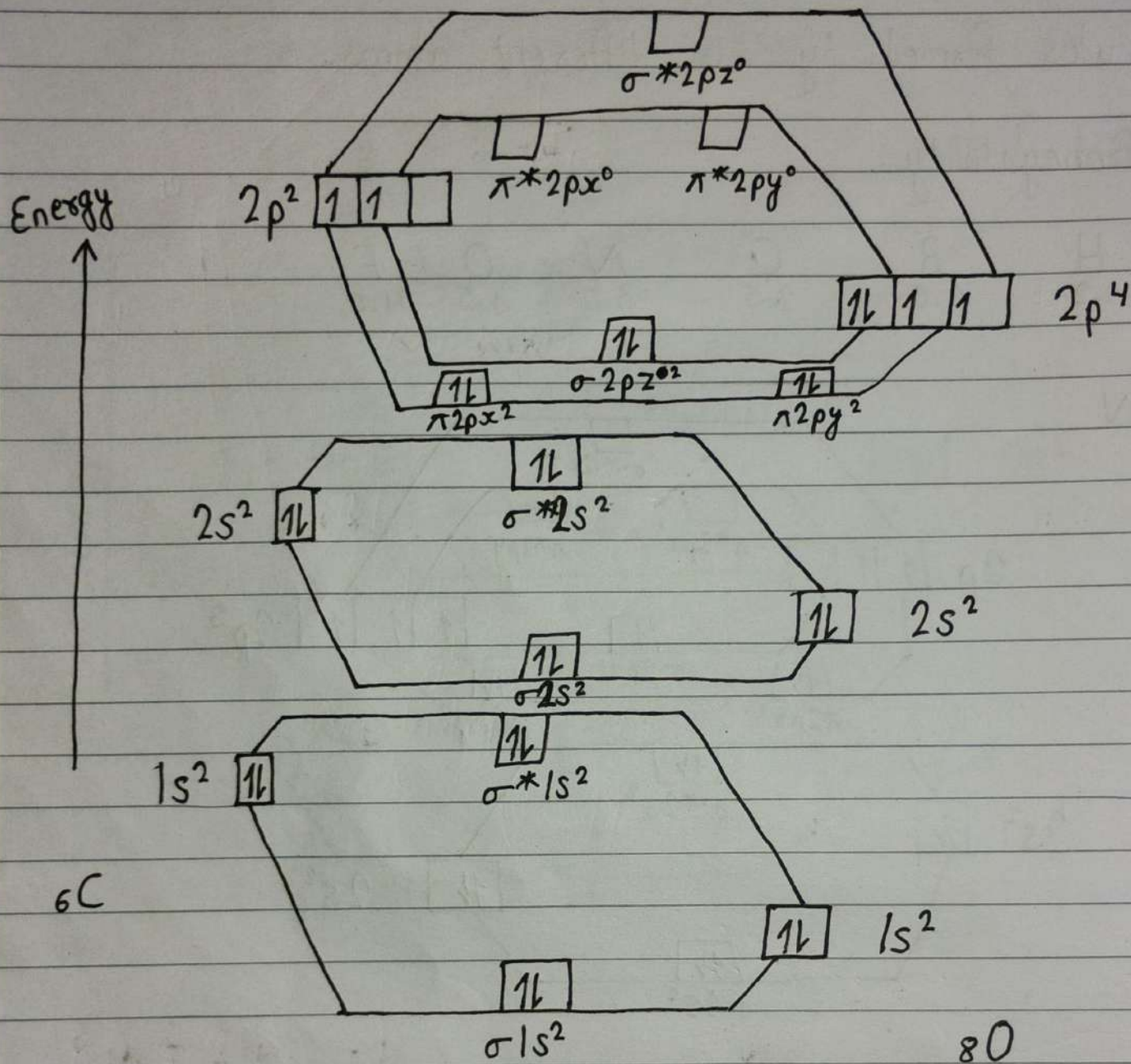
① CN



Configuration: $KK \sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \pi 2px^2 = \pi 2py^2 < \sigma 2pz^1 < \pi^* 2px^0 = \pi^* 2py^0 < \sigma^* 2pz^0$

$$B.O. = \frac{9-4}{2} = 2.5 \text{ (Paramagnetic)}$$

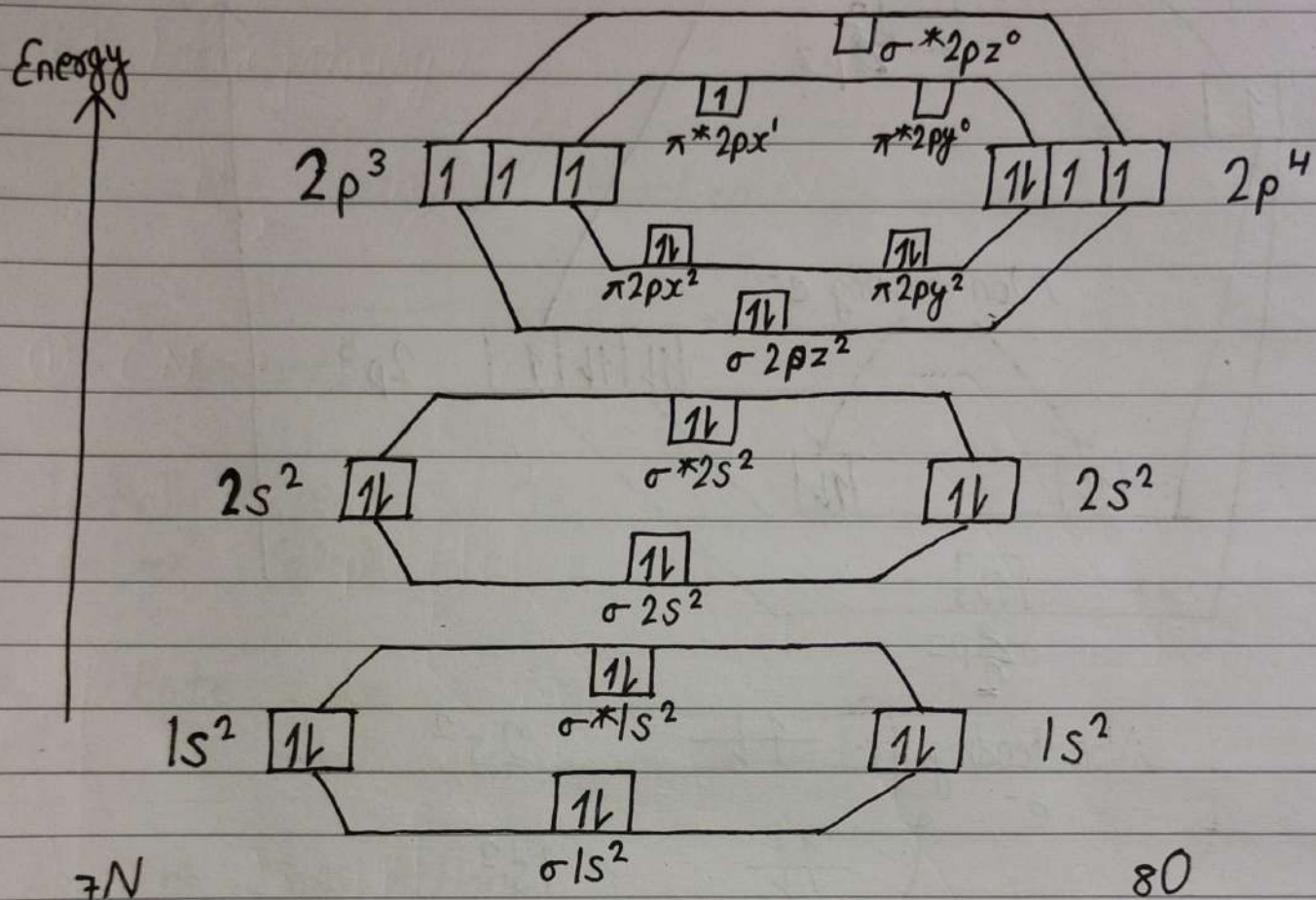
② CO



$$B.O. = \frac{10 - 4}{2} = 3$$

Magnetic behaviour = Diamagnetic.

③ NO

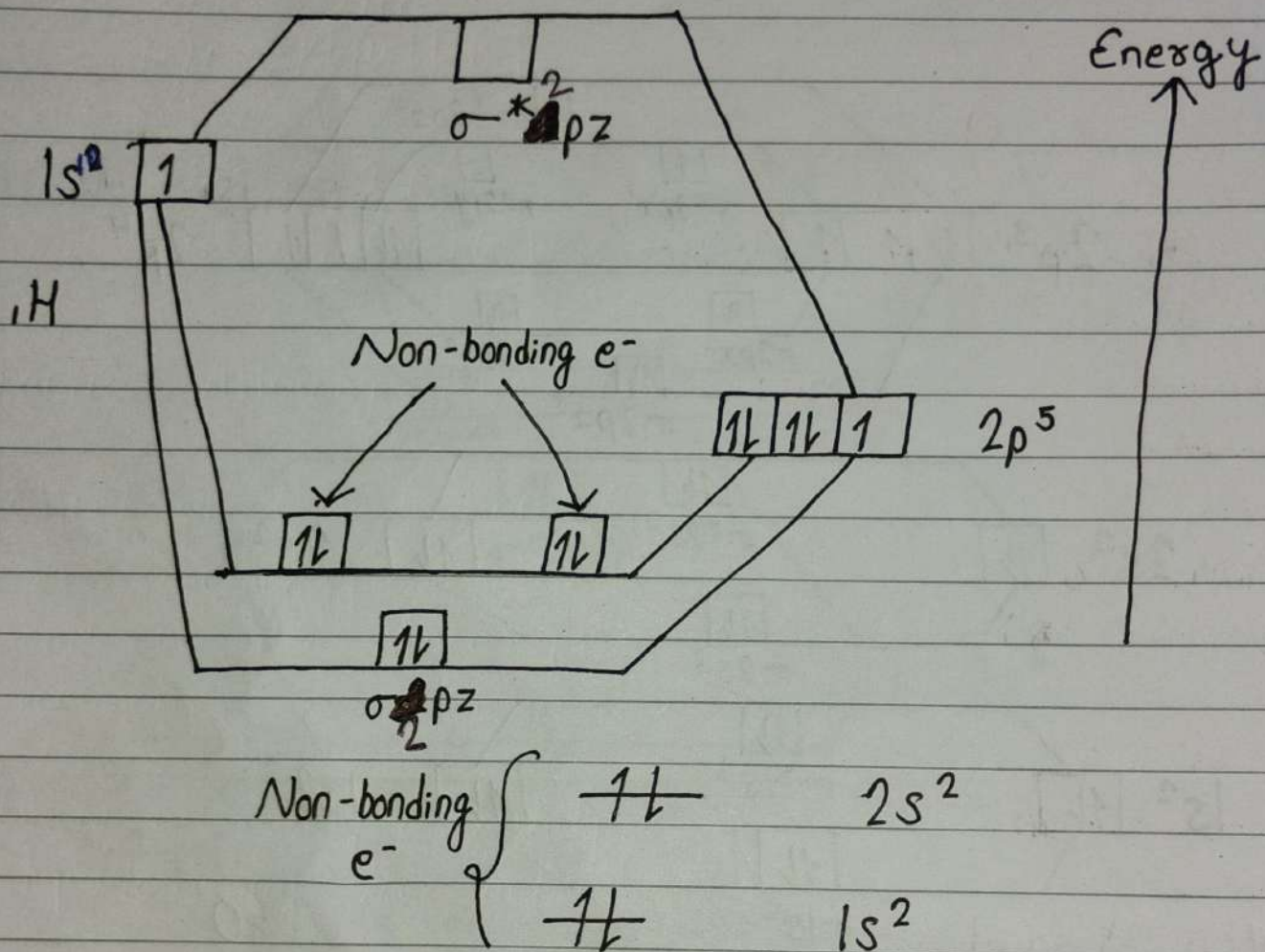


Configuration: $KK \sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \sigma 2p_z^2 < \pi 2p_x^2 = \pi 2p_y^2 < \pi^* 2p_x^1 = \pi^* 2p_y^0 < \sigma^* 2p_z^0$.

$$B.O. = \frac{10 - 5}{2} = 2.5$$

Magnetic behaviour = Paramagnetic.

Imp
④ HF



9F

$$B.O. = \frac{2-0}{2} = 1$$

Magnetic behaviour = Diamagnetic.